

Sustech 2025南科大入学考试数学样卷

NO. 1

The equation of a curve is $y = (2k - 3)x^2 - kx - (k - 2)$, where k is a constant. The line $y = 3x - 4$ is a tangent to the curve.

Find the value of k . [5]

NO. 2

Functions f and g are defined by

$$f : x \mapsto 3x - 2, \quad x \in \mathbb{R},$$

$$g : x \mapsto \frac{2x + 3}{x - 1}, \quad x \in \mathbb{R}, x \neq 1.$$

i. Obtain expressions for $f^{-1}(x)$ and $g^{-1}(x)$, stating the value of x for which $g^{-1}(x)$ is not defined. [4]

ii. Solve the equation $fg(x) = \frac{7}{3}$. [3]

NO. 3

Two points A and B have coordinates $(3a, -a)$ and $(-a, 2a)$ respectively, where a is a positive constant.

i. Find the equation of the line through the origin parallel to AB . [2]

ii. The length of the line AB is $3\frac{1}{3}$ units. Find the value of a . [3]

NO. 4

i. Prove the identity $\left(\frac{1}{\sin \theta} - \frac{1}{\tan \theta}\right)^2 \equiv \frac{1 - \cos \theta}{1 + \cos \theta}$. [3]

ii. Hence solve the equation $\left(\frac{1}{\sin \theta} - \frac{1}{\tan \theta}\right)^2 = \frac{2}{5}$, for $0^\circ \leq \theta \leq 360^\circ$. [4]

NO. 5

The coefficients of x and x^2 in the expansion of $(2 + ax)^7$ are equal. Find the value of the non-zero constant a . [3]

NO. 6

Relative to the origin O , the points A , B and C have position vectors given by

$$\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 3 \\ -2 \\ -4 \end{pmatrix}.$$

The quadrilateral $ABCD$ is a parallelogram.

a. Find the position vector of D . [3]

b. The angle between BA and BC is θ .

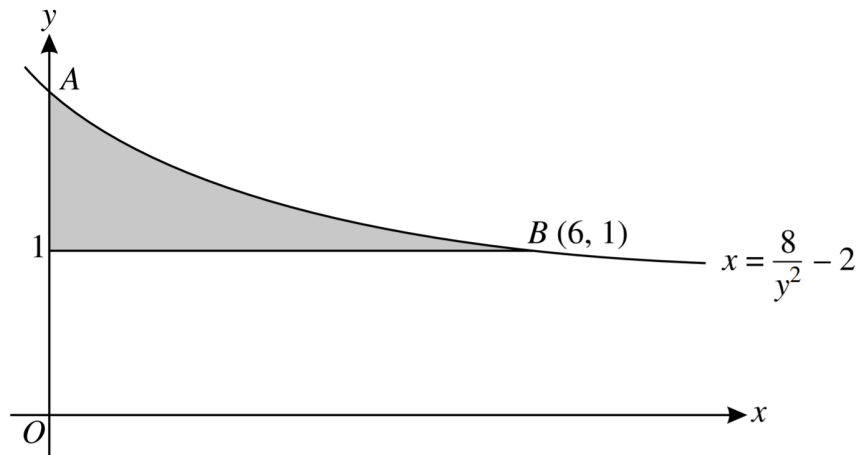
Find the exact value of $\cos \theta$. [3]

c. Hence find the area of $ABCD$, giving your answer in the form $p\sqrt{q}$, where p and q are integers. [4]

NO. 7

A curve has equation $y = 3x^3 - 6x^2 + 4x + 2$. Show that the gradient of the curve is never negative. [3]

NO. 8



The diagram shows part of the curve $x = \frac{8}{y^2} - 2$, crossing the y -axis at the point A. The point $B(6, 1)$ lies on the curve. The shaded region is bounded by the curve, the y -axis and the line $y = 1$. Find the exact volume obtained when this shaded region is rotated through 360° about the **y -axis**. [6]

NO. 9

Marco has four boxes labelled K , L , M and N . He places them in a straight line in the order K , L , M , N with K on the left. Marco also has four coloured marbles: one is red, one is green, one is white and one is yellow. He places a single marble in each box, at random. Events A and B are defined as follows.

A : The white marble is in either box L or box M .

B : The red marble is to the left of both the green marble and the yellow marble.

Determine whether or not events A and B are independent. [3]

NO. 10

Lengths of a certain type of white radish are normally distributed with mean μ cm and standard deviation σ cm. 4% of these radishes are longer than 12 cm and 32% are longer than 9 cm. Find μ and σ . [5]